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Assignment NO:- 5

Q.1] Count Digits in a Number

Input: n = 54321

Output: 5

Loop → kitivela ↳ while

n>0

cnt=0;

Steps	n%10	cnt++	n = n/10
1	54321%10 = 1	1	54321/10 = 5432
2	5432%10 = 2	2	5432/10 = 543
3	543%10 = 3	3	543/10 = 54
4	54%10 = 4	4	54/10 = 5
5	5%10 = 5	5	5/10 = 0

0>0 → false

```
int n = 54321;
int cnt=0; int digit=0;
while(n>0){
    digit = n%10;
        cnt++;
    n = n/10;
}
```

$\boxed{TC = O(\log_{10} n) \quad SC = O(1).}$

Q.2] Exact All Digit

input n = 5382

output = 2
8
3
5.

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n>0 int n = 5382

Steps. digit = n%10 digit n = n/10

1	= 5382%10 = 2	2	n = 5382/10 = 538
2	538%10 = 8	8	538/10 = 53
3	53%10 = 3	3	53/10 = 5
4	5%10 = 5	5	5/10 = 0
5	↓ n>0		

reverse condition.

int n = 5382;

```
while (n>0){
    int digit = n%10;
        s.o.p(digit);
        n = n/10;
}
```

n = n/10 → divide

$$\boxed{\begin{array}{l} TC = \log_{10} n \\ SC = O(1) \end{array}}$$

Q.3. Reverse a Number

Input: 12345

Output: 54321

n>0 int rev = 0;

Steps. rem = n%10 rev = rev*10 + rem; n = n/10.

1	12345%10 = 5	rev = 0*10 + 5 = 5	12345/10 = 1234
2	1234%10 = 4	= 5*10 + 4 = 54	1234/10 = 123
3	123%10 = 3	= 54*10 + 3 = 543	123/10 = 12
4	12%10 = 2	= 543*10 + 2 = 5432	12/10 = 1
5	1%10 = 1	= 5432*10 + 1 = 54321	1/10 = 0

condition false.

```
int rev = 0;
while(n>0){
    int rem = n%10;
    rev = rev*10 + rem;
    n = n/10;
} s.o.p(rev);
```

$$\boxed{\begin{array}{l} TC = \log n \\ SC = O(1) \end{array}}$$

Q.4. Palindrome Number

Input = 121

Output = Palindrome

Input = 123

Output = Not Palindrome

int rev=0; int num=n;

Steps	rem = n%10	rev=rev*10+rem	n = n/10;
1	= 121%10=1	rev=0*10+1 = 0+1=1	=121/10=12
2	12%10 = 2	rev=1*10+2 = 10+2=12	=12/10 = 1
3	1%10 = 1	rev=12*10+1 = 121	1/10 = 0

$\boxed{n=0}$ store num

num == rev ⇒ Palindrome

num != rev ⇒ Not Palindrome

~~int num=n; int~~

```
int n=121; int num=n; int rev=0;
while(n>0){
    int rem = n%10;
    rev=rev*10+rem;
    n=n/10;
}
if(num==rev){
    S.O.P("Palindrome");
}else{
    S.O.P("Not Palindrome");
}
```

$\boxed{TC = \log n \quad SC = O(1)}$

Q.5

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Q.5 Sum of Digits

Input = 543

Output = 12

5+4+3 = 12

int sum = 0;

Steps	digit = n%10	sum = sum+digit	n = n/10
1	543%10 = 3	= 0+3 = 3	= 543/10 = 54
2	54%10 = 4	= 3+4 = 7	54/10 = 5
3 (n>0)	5%10 = 5	= 7+5 = 12	5/10 = 0

Condition false.

```
int n = 543;
int sum = 0;
while(n>0){
    int digit = n%10;
    sum = sum+digit;
    n = n/10;
} s.o.p(sum);
```

$\boxed{\begin{array}{l} TC = \log n \\ SC = O(1) \end{array}}$

Q.6 Armstrong Number

Input = 153

$= 1^3 + 5^3 + 3^3 = 1 + 125 + 27 = 153$

Steps

- Count the digit
- count digit no is Power each Extract no.
- ~~After th~~ add the number
- check if sum equal to given no.

int n = 153; int num = n; int sum = 0;

cnt = count the Power cnt = 0;

Steps	rem = n%10	cnt++	n = n/10
1	= 153%10 = 3	1	= 153/10 = 15
2	15%10 = 5	2	15/10 = 1
3	1%10 = 1	3	1/10 = 0

$\boxed{cnt = 3}$

153%10

~~digit = n%10~~

~~1 ≤ 3~~

~~int i=1 i ≤ cnt i++~~

~~1 + 25 + 27~~

int power = 1; int sum = 0; n = num;

digit = n % 10 n = ~~n/~~ n = n/10;

= 153 % 10 = 3

↳ for(int i=1; i ≤ cnt; i++)

↳ 1 ≤ 3

↳ Power = Power * digit

= 3 * 1 = 3

→ 2 ≤ 3

↳ = 3 * 3 = 9

→ 3 ≤ 3

↳ 9 * 3 = 27

→ 4 ≤ 3 ×

→ sum = Power + sum

= 27 + 0 = 27

Power = 1;

= 153/10 = 15

digit = 15 % 10 = 5

↳ for(int i=1; i ≤ cnt; i++)

1 ≤ 3

↳ Power = Power * digit

= 1 * 5 = 5

→ 2 ≤ 3

↳ Power = Power * digit

= 5 * 5 = 25

→ 3 ≤ 3

↳ Power = Power * digit

= 25 * 5 = 125

→ 4 ≤ 3 ×

→ Sum = Power + Sum

= 125 + 27 = 152.

Power = 1;

15/10 = 1

digit = 1 % 10 = 1

i = cnt, i++

→ 1 ≤ 3

↳ Power = Power * digit

= 1 * 1 = 1

→ 2 ≤ 3

↳ = 1 * 1 = 1

→ 3 ≤ 3 = 1 * 1 = 1

↳ 4 ≤ 3 ×

→ sum = Power + sum

= 1 + 152 = 153

Power = 1

n = n/10

= 1/10 = 0.

0 > 0 → condition false

if num == sum → Armstrong number

num != sum → Not Armstrong number

int n = 153; int num = n; int cnt = 0; int sum = 0;

→ [n, 1]

```
while(n>0){
    int rem = n%10;
    cnt++;
    n = n/10;
}
```

$\boxed{\begin{array}{l} TC = O(n) \\ SC = O(1) \end{array}}$

~~while~~ ~~int power = 1;~~ n = num; (1, n)

```
while(n>0){
    int digit = n%10; int power = 1;
    for(int i = 1; i ≤ cnt; i++){ → [1, 3]
        power = power * digit;
    }
    sum = power + sum;
}
```

O(n) * 3 = O(3n) * O(n)

= O(4n)

= O(n)

```
~~s.o.p~~ if(num == sum){
        s.o.p("Armstrong No");
} else {
        s.o.p("Not Armstrong No");
}
```

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Q.7 Print All Divisors $n = 12$

~~divisor~~ if $n \% i == 0$ → condition in if

divisor. i++.

$i = 1$

$= 12 \% 1 = 0$

↳ 1

2

$12 \% 2 = 0$

↳ 2

3

$12 \% 3 = 0$

↳ 3

4

$12 \% 4 = 0$

↳ 4

5

$12 \% 5 = 0$ ×

6

$12 \% 6 = 0$

↳ 6

7

$12 \% 7 =$ ×

8

$12 \% 8 =$ ×

9

$12 \% 9$ ×

10

$12 \% 10$ ×

11

$12 \% 11$ ×

12

$12 \% 12$

↳ 12

Output = 1 2 3 4 6 12.

```
int n=12;
for(int i=1; i≤n; i++){
    if(n%i==0){
        S.O.P(i);
    }
}
```

$\boxed{\begin{array}{l} TC = O(n) \\ SC = O(1) \end{array}}$

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Q.8 Count Number of Divisors.

int n = 12; int cnt = 0;

for int i = 1 i ≤ n

i++

→ 1 ≤ 12
↳ 12 % 1 == 0
↳ 1

2

→ 2 ≤ 12
↳ 12 % 2 == 0
↳ 2

3

→ 3 ≤ 12
↳ 12 % 3 == 0
↳ 3

4

→ 4 ≤ 12
↳ 12 % 4 == 0
↳ 4

5

→ 5 ≤ 12
↳ 12 % 5 == 0 ✗

6

→ 6 ≤ 12
↳ 12 % 6 == 0
↳ 5

7

→ 7 ≤ 12
↳ 12 % 7 == 0 ✗

8

→ 8 ≤ 12
↳ 12 % 8 == 0 ✗

9

→ 9 ≤ 12
↳ 12 % 9 == 0 ✗

10

→ 10 ≤ 12
↳ 12 % 10 == 0 ✗

11

→ 11 ≤ 12
↳ 12 % 11 == 0 ✗

12

→ 12 ≤ 12
↳ 12 % 12 == 0
↳ 6

13

13 ≤ 12 ✗

Output = 6

```
int n=12;  int cnt=0;
for(int i=1; i≤n; i++){
        if(n%i==0){
                cnt++;
        }
}
s.o.p(cnt);
```

Q.9 check prime Number.

Input: 7

Output = prime.

```
int n=7;
```

~~i=1~~ ~~i≤n~~

↳ i≤7

n=7 ↳

7%1 = 0 → 1

7%2 = 1 ×

7%3 = 1 ×

7%4 = 3 ×

7%5 = 2 ×

7%6 = 1 ×

7%7 = 0 → 2

↳ Prime = 2 divisor

n = 4

4%1 = 0 → 1

4%2 = 0 → 2

4%3 = 1

4%4 = 0 → 3

↳ not prime

↳ 3 divisors

```
int n=7;  int cnt=0;
for(int i=1; i≤n; i++){
        if(n%i==0){
                cnt++;
        }
}
if(cnt==2){
        s.o.p("Prime");
}else{
        s.o.p("not Prime")}
}
```

TC = O(n)
SC = O(1)

(Q.10) Print All prime numbers from 1 to n.

Input: n = 20.

Output:

2 3 5 7 11 13 17 19

1 → 1%1 → 0 → complete divisor → 1

2 → 2%1 → 0 2%2 → 0 → divisor → 2

3 → 3%1 → 0 3%2 → 1 3%3 → 0 → divisor → 2

4 → 4%1 → 0 4%2 → 0 4%3 → 1 4%4 → 0 → divisor → 3

5 → 5%1 → 0 5%2 → 1 5%3 → 2 5%4 → 1 5%5 → 0 → divisor → 2

6 → 6%1 → 0 6%2 → 0 6%3 → 0 6%4 → 2 6%5 → 1 6%6 → 0 → divisor → 4

7 → 7%1 → 0 7%2 → 1 7%3 → 1 7%4 → 3 7%5 → 2 7%6 → 1 7%7 → 0 complete divisor → 2

8 → 8%1 → 0 8%2 → 0 8%3 → 2 8%4 → 0 8%5 → 3 8%6 → 2 8%7 → 1 8%8 → 0. complete divisor → 4.

9 = 9%1 → 0 9%2 → 1 9%3 → 0 9%4 → 1 9%5 → 4 9%6 → 3 9%7 → 2 9%8 → 1 9%9 → 0.

divisor → 3.

```
int n = 20;
for(int i = 1; i ≤ n; i++){ → (1, n)
    int cnt = 0;
    for(int j = 1; j ≤ i; j++){
        if(i%j == 0){
            cnt++;
        }
    }
    ~~if(cnt == 2){ s.o~~ if(cnt == 2){
        s.o.p(i);
    }
}
```

$\boxed{TC = O(n^2) \quad SC = O(1)}$

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Q.11] GCD find.

Input:

a = 12

b = 18

output = 6

Steps → 1) small number find out

2) for (1 — minnumber)

3] complete divisor no & find greater no.

```
int min=0; int gcd=1;
if(a>b){
    min=b;
}else{
    min=a;
}
for(int i=1; i≤min; i++){
    if(min%i==0){
        gcd=i;
    }
s.o.p(gcd);
```

$\boxed{TC = O(\min(a,b)) \quad SC = O(1)}$

a > b = 12>18 → false → else → min = 12.

int i=1

i ≤ 12

1 ≤ 12

↳ 12%1 == 0

↳ gcd = 1

i++

2

2 ≤ 12

↳ 12%2 == 0

↳ gcd = 2

3

3 ≤ 12

↳ 12%3 == 0

↳ gcd = 3

4

4 ≤ 12

↳ 12%4 == 0

↳ gcd = 4

5 ≤ 12

~~↳ 12%5 == 2~~

6 ≤ 12

↳ 12%6 == 0

↳ gcd = 6

7 ≤ 12

↳ 12%7 == 5

8 ≤ 12

↳ 12%8 == 4

9 ≤ 12

↳ 12%9 == 3

10 ≤ 12

↳ 12%10 = 2

11 ≤ 12

↳ 12%11 == 1

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Q.12] GCD using Euclidean Algorithm

Input:

a = 48

b = 18.

$gcd(a,b) = gcd(a\%b, b)$, $a > b$

$(48\%18, 18) = (12, 18) = (12, 18\%12)$

$= (12, 6) = (12\%6, 6)$

$= (0, 6)$

a b

$a == 0 \Rightarrow gcd = b$

$a != 0 \Rightarrow gcd = a$

```
int a=48; int b=18; ~~int min=0;~~ int gcd=1
~~if (a>b){~~
~~    min=b;~~
~~}else{~~
~~    min=a;~~
~~}~~
~~for (int i=1;~~
while(a!=0 && b!=0){
    if(a>b){
        a=a%b;
    }else{
        b=b%a
    }
}
if(a==0){
    gcd=b;
}else{
    gcd=a;
}
```

$\boxed{\begin{array}{l} SC = O(1) \\ TC = (\log \text{maximum}(a,b)) \end{array}}$

Q.13. Check coprime Numbers

Input:

8 15

Output: coprime

Divisors of 8: 1, 2, 4, 8

Divisors of 15: 1, 3, 5, 15

gcd = 1

o.p: co-prime.

(10, 12)

Divisors of 10: 1, 2, 5, 10

Divisors of 12: 1, 2, 3, 4, 6, 12.

gcd = 2

o.p: Not Co-prime

$$\begin{array}{r} 1 \\ 8\overline{)15} \\ -8 \\ \hline 7 \end{array} \qquad \begin{array}{r} 7 \\ 1\overline{)7} \\ -7 \\ \hline 0 \end{array}$$

Algorithm

if((8,15) = (8, 15%8) = (8, 7) , (8%7, 7) = (1, 7) = (1, 7%1)

(r, 0)

```
int a = 8;  int b = 15;

while (a>0 && b>0){
    if(a>b){
        a = a%b;
    }else{
        b = b%a;
    }
}
if(a==0){
    gcd = b;
}else{
    gcd = a;
}
```

```
if (gcd==1){
    s.o.p("coprime");
}else{
    s.o.p("Not coprime");
}
```

TC = (log(Max(min(a,b))))

SC = O(1)

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Q.14] Check perfect Number

Sum of its Proper divisors = number

Input = 6

Divisors: 1 2 3

1 + 2 + 3 = 6

OP: Perfect

Input = 8

Divisors: 1, 2, 4,

1+2+4 = 7.

OP = Not perfect

n = 6

Div = 6%1

Div = 0

Div = 6%2 = 0

Div = 6%3 = 0

Div = 6%4 = 2

Div = 6%5 = 1

6 > 6 = false

if(Div==0){

Sum=sum+1

= 0+1 = 1

↳ = 1+2 = 3

↳ = 3+3 = 6

↳ x

↳ x

~~Sum = 6. n =~~ sum == n

6 == 6 → Perfect

n = 8.

Div = 8%1 = 0

Div = 8%2 = 0

Div = 8%3 = 2

Div = 8%4 = 0

Div = 8%5 = 3

Div = 8%6 = 2

Div = 8%7 = 1

8 < 8 = false

if(Div==0){

↳ sum = sum+1

= 0+1 = 1

↳ 1+2 = 3

↳ x

↳ 3+4 = 7

↳ x

↳ x

↳ x

sum == n

↳ 7 == 8 → Not Perfect

```
int n=6; ~~int Diviso~~ int sum=0;
for (int i=1; i<n; i++){ -> (n/n)
    int Divisor = n%i
    if (Divisor ==0){
        sum = sum+i;
    }
}
if(sum ==n){
    S.O.P("Perfect No");
}else{
    S.O.P("Not Perfect No");
}
```

$\boxed{\begin{array}{l} TC = O(n) \\ SC = O(1) \end{array}}$